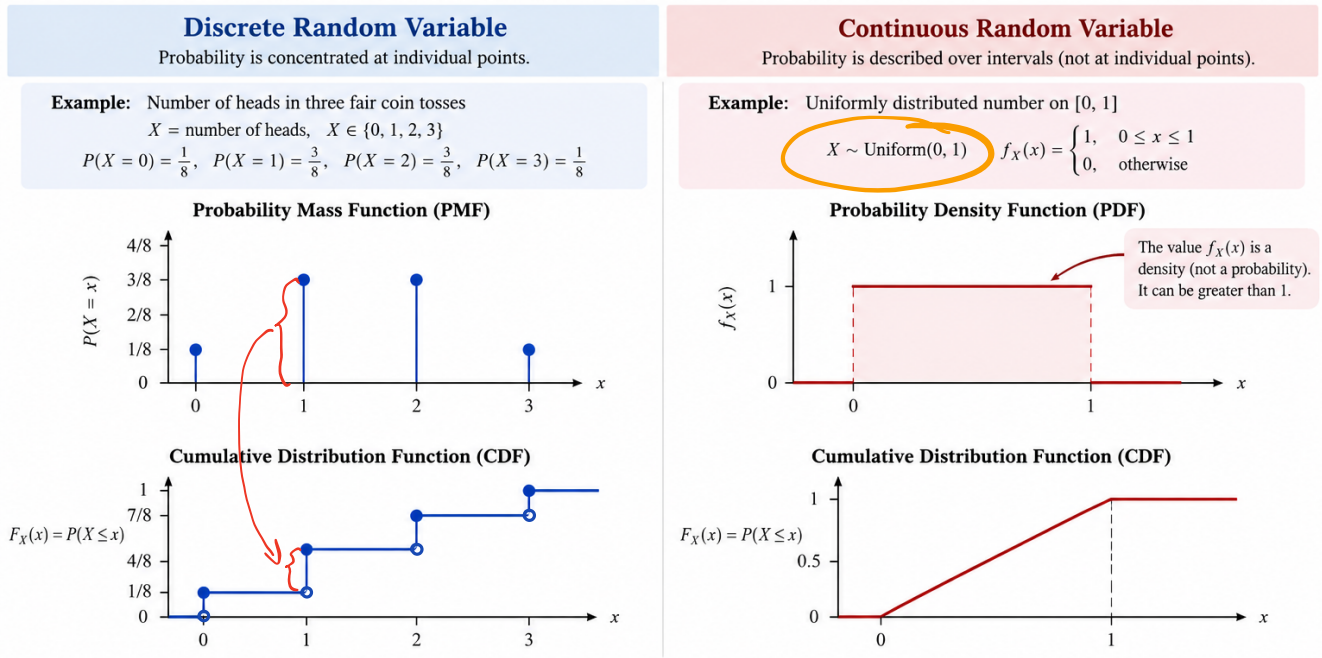
Last Time



	Discrete	Continuous
Possible values	Countable (e.g., $\{0, 1, 2, 3\}$)	Uncountable (e.g., all real numbers in an interval)
Distribution	Probability mass function $p_X(x) = P(X = x)$	Probability density function $f_X(x)$ (not a probability)
Probability at a point	Can be positive (e.g., $P(X = 1) = \frac{3}{8}$)	Zero (e.g., $P(X = x) = 0$ for any x)
Probability over a set	Sum: $P(a \le X \le b) = \sum_{x \in [a,b]} p_X(x)$	Integral: $P(a \le X \le b) = \int_a^b f_X(x)\,dx$
CDF shape	Step function (jumps at points)	Continuous function (no jumps)

* This figure was generated by ChatGPT based on the instructor's instructions and verified by the instructor.

Recall:

Continuous Random Variable & Probability Density Function

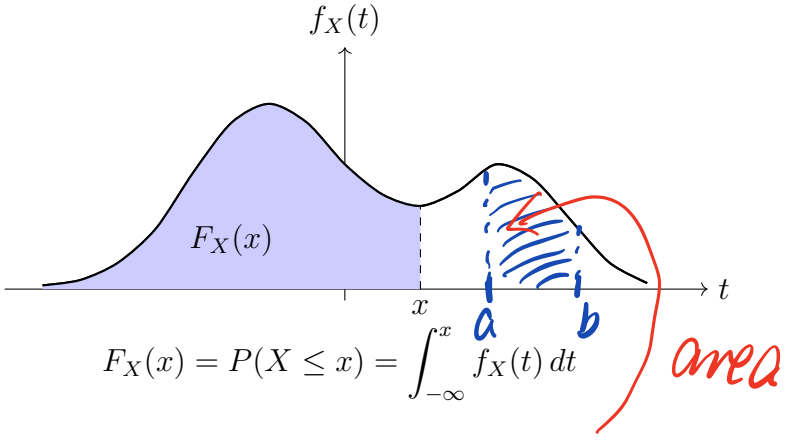
A random variable X is called a ***continuous random variable*** if there exists a nonnegative function

$$f_X : \mathbb{R} \to [0, \infty)$$

such that, for any $a < b$,

$$P(a < X \le b) = \int_a^b f_X(x)\,dx.$$

The function f_X is called the ***probability density function (PDF)*** of X.



$$P(a < X \le b) = \int_a^b f_X(x)\,dx$$

Families of Continuous Random Variables

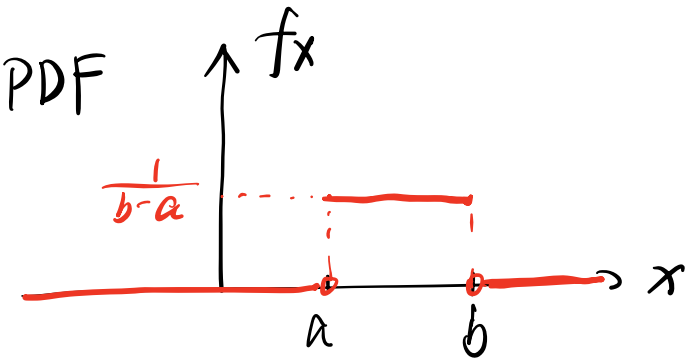
Uniform Random Variable

X is a uniform (a, b) random variable if the probability density function (PDF) of X is

$$f_X(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x \le b, \\ 0, & \text{otherwise.} \end{cases}$$

where the two parameters are $b > a$.

Sketch both PDF and CDF of X.



why $\frac{1}{b-a}$?

$\int_{-\infty}^{\infty} f_X(x)\,dx = 1$

$\therefore$ The constant must be $\frac{1}{b-a}$

CDF:

$$F_X(x) = \int_{-\infty}^{x} f_X(u)\,du$$

when $x < a$, we have

$$F_X(x) = \int_{-\infty}^{x} f_X(u)\,du = \int_{-\infty}^{x} 0\,du = 0$$

when $a \le x \le b$, we h

$$F_X(x) = \int_{-\infty}^{x} f_X(u)\,du = \int_{-\infty}^{a} f_X(u)\,du + \int_{a}^{x} f(u)\,du$$

$$= 0 + \int_a^x \frac{1}{b-a}\, du$$

$$= \frac{1}{b-a} \cdot (x-a)$$

when $x > b$, we have

$$F_X(x) = \int_{-\infty}^{x} f_X(u)\, du$$

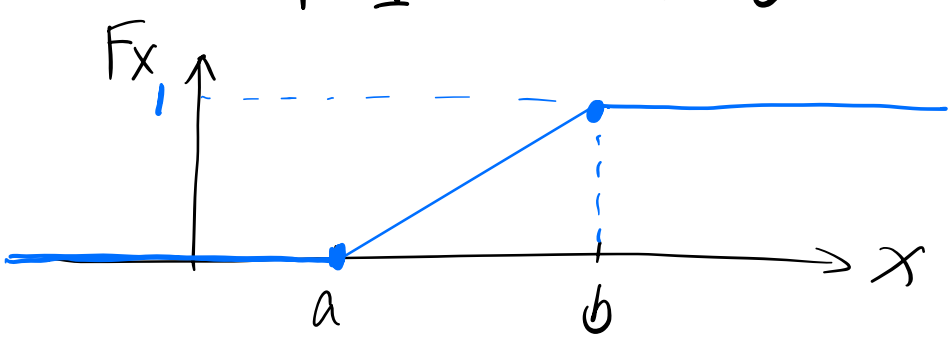
$$= \int_{-\infty}^{a} f_X(u)\, du + \int_a^b f_X(u)\, du + \int_b^x f_X(u)\, du$$

$$= 0 + \frac{b-a}{b-a} + 0$$

$$= 1$$

$\therefore$

$$F_X(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a \le x \le b \\ 1, & x > b \end{cases}$$



Exponential Random Variable

X is an exponential (λ) random variable if the probability density function (PDF) of X is

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

where the parameter $\lambda > 0$.

Example [Component Lifetime] The lifetime X (in hours) of an electronic component is modeled as an *exponential random variable* with parameter $\lambda = 0.002$. Thus, its PDF is

$$f_X(x) = \begin{cases} 0.002e^{-0.002x}, & x \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

Find the probability that the component operates for at least 500 hours before failing.

$P(X \geq 500) = 1 - P(X < 500)$

$\left(= \int_{-\infty}^{500} f_X(x)\,dx\right)$

$= 1 - \int_{-\infty}^{500} f_X(x)\,dx$

$= \int_{-\infty}^{\infty} f_X(x)\,dx - \int_{-\infty}^{500} f_X(x)\,dx$

$= \int_{500}^{\infty} f_X(x)\,dx$

$= \int_{500}^{\infty} 0.002\,e^{-0.002x}\,dx$

$= \left[-e^{-0.002x}\right]\Big|_{500}^{\infty} = e^{-1} \approx 0.3679.$

Gaussian (normal) random variables are fundamental in engineering because they naturally model noise, measurement errors, and the combined effects of many small random variations encountered in real-world systems.

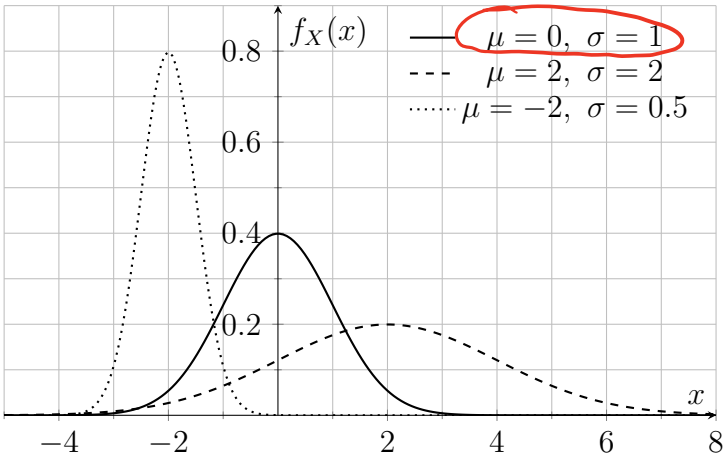
Gaussian (normal) Random Variable

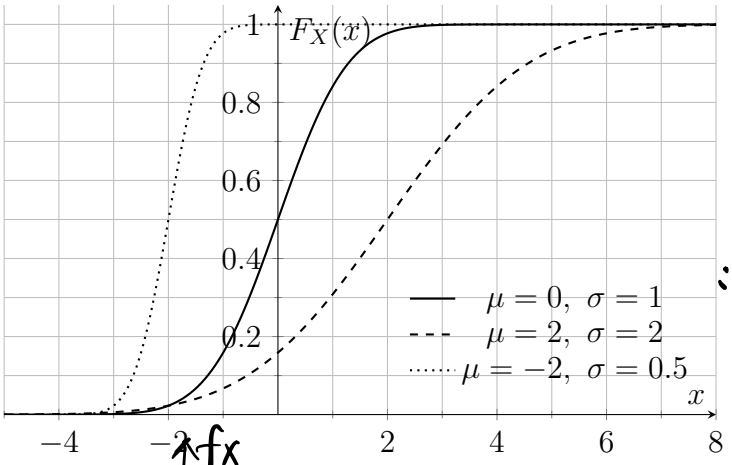
X is a ***Gaussian random variable with parameters*** μ ***and*** σ if the probability density function (PDF) of X is

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2},$$

where the parameter μ can be any real number and the parameter $\sigma > 0$.

- We denote $X \sim N(\mu, \sigma^2)$.
- The graph of $f_X(x)$ has a bell shape:
  - the center of the bell is $x = \mu$;
  - σ reflects the width of the bell
    - if σ is small, the bell is narrower with a higher peak;
    - if σ is large, the bell is wider with a lower peak.





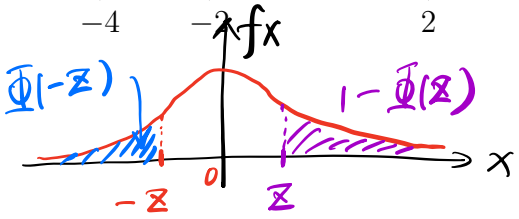
If $Z \sim N(0,1)$, then Z is called the standard normal random variable, and we denote its CDF as

$$\Phi(z) := F_Z(z) = \int_{-\infty}^{z} f_X(x)\,dx$$

$$\therefore \Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-x^2/2}\,dx$$

and we also have

$$\Phi(-z) = 1 - \Phi(z)$$



If $X \sim N(\mu, \sigma^2)$, then

$$F_X(x) = \int_{-\infty}^{x} f_X(w)\,dw = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(w-\mu)^2/2\sigma^2}\,dw$$

change variable

$$s = \frac{w-\mu}{\sigma}$$

$$= \int_{-\infty}^{\frac{x-\mu}{\sigma}} \frac{1}{\sqrt{2\pi}} e^{-s^2/2}\,ds$$

$$= \Phi\left(\frac{x-\mu}{\sigma}\right)$$

The standard normal CDF $\Phi(z)$

z	$\Phi(z)$	z	$\Phi(z)$	z	$\Phi(z)$	z	$\Phi(z)$	z	$\Phi(z)$	z	$\Phi(z)$
0.00	0.5000	0.50	0.6915	1.00	0.8413	1.50	0.9332	2.00	0.97725	2.50	0.99379
0.01	0.5040	0.51	0.6950	1.01	0.8438	1.51	0.9345	2.01	0.97778	2.51	0.99396
0.02	0.5080	0.52	0.6985	1.02	0.8461	1.52	0.9357	2.02	0.97831	2.52	0.99413
0.03	0.5120	0.53	0.7019	1.03	0.8485	1.53	0.9370	2.03	0.97882	2.53	0.99430
0.04	0.5160	0.54	0.7054	1.04	0.8508	1.54	0.9382	2.04	0.97932	2.54	0.99446
0.05	0.5199	0.55	0.7088	1.05	0.8531	1.55	0.9394	2.05	0.97982	2.55	0.99461
0.06	0.5239	0.56	0.7123	1.06	0.8554	1.56	0.9406	2.06	0.98030	2.56	0.99477
0.07	0.5279	0.57	0.7157	1.07	0.8577	1.57	0.9418	2.07	0.98077	2.57	0.99492
0.08	0.5319	0.58	0.7190	1.08	0.8599	1.58	0.9429	2.08	0.98124	2.58	0.99506
0.09	0.5359	0.59	0.7224	1.09	0.8621	1.59	0.9441	2.09	0.98169	2.59	0.99520
0.10	0.5398	0.60	0.7257	1.10	0.8643	1.60	0.9452	2.10	0.98214	2.60	0.99534
0.11	0.5438	0.61	0.7291	1.11	0.8665	1.61	0.9463	2.11	0.98257	2.61	0.99547
0.12	0.5478	0.62	0.7324	1.12	0.8686	1.62	0.9474	2.12	0.98300	2.62	0.99560
0.13	0.5517	0.63	0.7357	1.13	0.8708	1.63	0.9484	2.13	0.98341	2.63	0.99573
0.14	0.5557	0.64	0.7389	1.14	0.8729	1.64	0.9495	2.14	0.98382	2.64	0.99585
0.15	0.5596	0.65	0.7422	1.15	0.8749	1.65	0.9505	2.15	0.98422	2.65	0.99598
0.16	0.5636	0.66	0.7454	1.16	0.8770	1.66	0.9515	2.16	0.98461	2.66	0.99609
0.17	0.5675	0.67	0.7486	1.17	0.8790	1.67	0.9525	2.17	0.98500	2.67	0.99621
0.18	0.5714	0.68	0.7517	1.18	0.8810	1.68	0.9535	2.18	0.98537	2.68	0.99632
0.19	0.5753	0.69	0.7549	1.19	0.8830	1.69	0.9545	2.19	0.98574	2.69	0.99643
0.20	0.5793	0.70	0.7580	1.20	0.8849	1.70	0.9554	2.20	0.98610	2.70	0.99653
0.21	0.5832	0.71	0.7611	1.21	0.8869	1.71	0.9564	2.21	0.98645	2.71	0.99664
0.22	0.5871	0.72	0.7642	1.22	0.8888	1.72	0.9573	2.22	0.98679	2.72	0.99674
0.23	0.5910	0.73	0.7673	1.23	0.8907	1.73	0.9582	2.23	0.98713	2.73	0.99683
0.24	0.5948	0.74	0.7704	1.24	0.8925	1.74	0.9591	2.24	0.98745	2.74	0.99693
0.25	0.5987	0.75	0.7734	1.25	0.8944	1.75	0.9599	2.25	0.98778	2.75	0.99702
0.26	0.6026	0.76	0.7764	1.26	0.8962	1.76	0.9608	2.26	0.98809	2.76	0.99711
0.27	0.6064	0.77	0.7794	1.27	0.8980	1.77	0.9616	2.27	0.98840	2.77	0.99720
0.28	0.6103	0.78	0.7823	1.28	0.8997	1.78	0.9625	2.28	0.98870	2.78	0.99728
0.29	0.6141	0.79	0.7852	1.29	0.9015	1.79	0.9633	2.29	0.98899	2.79	0.99736
0.30	0.6179	0.80	0.7881	1.30	0.9032	1.80	0.9641	2.30	0.98928	2.80	0.99744
0.31	0.6217	0.81	0.7910	1.31	0.9049	1.81	0.9649	2.31	0.98956	2.81	0.99752
0.32	0.6255	0.82	0.7939	1.32	0.9066	1.82	0.9656	2.32	0.98983	2.82	0.99760
0.33	0.6293	0.83	0.7967	1.33	0.9082	1.83	0.9664	2.33	0.99010	2.83	0.99767
0.34	0.6331	0.84	0.7995	1.34	0.9099	1.84	0.9671	2.34	0.99036	2.84	0.99774
0.35	0.6368	0.85	0.8023	1.35	0.9115	1.85	0.9678	2.35	0.99061	2.85	0.99781
0.36	0.6406	0.86	0.8051	1.36	0.9131	1.86	0.9686	2.36	0.99086	2.86	0.99788
0.37	0.6443	0.87	0.8078	1.37	0.9147	1.87	0.9693	2.37	0.99111	2.87	0.99795
0.38	0.6480	0.88	0.8106	1.38	0.9162	1.88	0.9699	2.38	0.99134	2.88	0.99801
0.39	0.6517	0.89	0.8133	1.39	0.9177	1.89	0.9706	2.39	0.99158	2.89	0.99807
0.40	0.6554	0.90	0.8159	1.40	0.9192	1.90	0.9713	2.40	0.99180	2.90	0.99813
0.41	0.6591	0.91	0.8186	1.41	0.9207	1.91	0.9719	2.41	0.99202	2.91	0.99819
0.42	0.6628	0.92	0.8212	1.42	0.9222	1.92	0.9726	2.42	0.99224	2.92	0.99825
0.43	0.6664	0.93	0.8238	1.43	0.9236	1.93	0.9732	2.43	0.99245	2.93	0.99831
0.44	0.6700	0.94	0.8264	1.44	0.9251	1.94	0.9738	2.44	0.99266	2.94	0.99836
0.45	0.6736	0.95	0.8289	1.45	0.9265	1.95	0.9744	2.45	0.99286	2.95	0.99841
0.46	0.6772	0.96	0.8315	1.46	0.9279	1.96	0.9750	2.46	0.99305	2.96	0.99846
0.47	0.6808	0.97	0.8340	1.47	0.9292	1.97	0.9756	2.47	0.99324	2.97	0.99851
0.48	0.6844	0.98	0.8365	1.48	0.9306	1.98	0.9761	2.48	0.99343	2.98	0.99856
0.49	0.6879	0.99	0.8389	1.49	0.9319	1.99	0.9767	2.49	0.99361	2.99	0.99861

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}$$

Example [Sensor Measurement Error] A temperature sensor has measurement error X (in °C) modeled as a Gaussian random variable with

$$\mu = 0, \qquad \sigma = 0.5.$$

Thus, its PDF is

$$f_X(x) = \frac{1}{0.5\sqrt{2\pi}} \exp\left(-\frac{x^2}{2(0.5)^2}\right).$$

The sensor is considered sufficiently accurate if the magnitude of its measurement error is less than 0.8°C. Find the probability that a measurement is sufficiently accurate.

Mixed Random Variables

Mixed Random Variable

A random variable X is of **mixed type** if its distribution contains both a discrete component and a continuous component.

In other words, there is at least one value x such that $P(X = x) > 0$ and the sum of the probabilities of all values x with positive probability is not one.

Example [Received Signal Level] Suppose X represents the received signal level at a wireless receiver. Due to complete signal loss,

$$P(X = 0) = 0.2.$$

If the signal is successfully received, which occurs with probability 0.8, suppose its level is uniformly distributed over $(0, 10)$. Thus, X has both a discrete component and a continuous component.

(a) Find the probability that the received signal level is between 2 and 5.

(b) Sketch the CDF of X.